DEEP LEARNING FOR UNSUPERVISED DENOISING OF RADAR SOUNDER DATA

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ABSTRACT

Analyzing radar sounder (RS) profiles allows the retrieval of critical information on subsurface geology. However, radargrams suffer from several noise contributions, adversely affecting the data quality and reliability. In the remote sensing literature, there are no methods for denoising radargrams, and those for optical data denoising and SAR and GPR data despeckling are based on assumptions that are not valid in the RS domain. This paper analyses the statistical distributions of the noisy contributions in radargrams at different levels of processing. It proposes a novel method to denoise complex raw radargrams using a generative network (diffusion probabilistic model) that learns the noise statistical properties. The model is iteratively trained to learn the information loss as a function of the noise level increment in the data. By reversing the process, the network estimates the noise statistical properties and denoises unseen radargrams. The method has been successfully validated on the raw Experiment Data Record (EDR) radargrams of Mars that the Shallow Radar (SHARAD) acquired.

***Index Terms*—** denoising, despeckling, deep learning, denoising diffusion probabilistic models, radar sounder.

1. INTRODUCTION

Radar sounder (RS) profiles, also known as radargrams, have been unveiling unique geophysics information on the subsurface geology for planetary exploration and Earth observation to investigate subsurface structures, mapping geological features, and detecting underground targets [1]. However, radargrams suffer from several noisy contributions, adversely affecting the quality and reliability of the data and making accurate interpretation and analysis challenging.

In the remote sensing literature, there are no methods for denoising radargrams but rather methods for RS data resolution enhancement, interference suppression, and filtering. Numerous techniques have been developed for denoising optical, ground penetrating radar (GPR), and synthetic-aperture radar (SAR) data based on assumptions that are not valid for radargrams. Denoising methods for optical data assume the noise to be additive and Gaussian distributed, while radargrams also contain speckle noise that can be approximated as multiplicative. Methods for GPR and SAR denoising (mostly despeckling) address the problem of multiplicative noise by exploiting i) couples of SAR data and noise-free data (supervised methods) or ii) multitemporal data acquired over the same track, assuming they contain the same signal and independent and identically distributed (i.i.d.) speckle noise. These methods cannot be applied to RS data since multitemporal data acquired over the same track and/or noise-free radargrams do not exist. Further, these methods assume that the signal covers most of the image. In contrast, RS data are characterized by large non-backscattering areas with extremely different statistical properties than the signal [2]. Classical remote sensing denoising methods exploit various filtering, transform, and decomposition techniques that require user interaction. Advanced GPR and SAR data denoising and despeckling techniques are mostly based on deep learning (e.g., [3]), given its ability to extract information from the data automatically. Hence, a need emerges for novel radargram denoising methods considering RS data-specific statistical properties.

Despite the lack of methods for denoising radargrams, deep learning has been applied to radargrams obtaining good performance for i) ice layers and basal interface detection, ii) radargram simulation, iii) unsupervised target detection [4], iv) supervised [5, 6] and unsupervised [7] segmentation, v) ice bed detection, and vi) radargram super-resolution. Given the deep learning ability to handle the complexity of RS data and to model challenging tasks in the RS domain, deep learning looks promising for denoising and despeckling RS data.

This paper analyses the statistical distributions of the noisy contributions in radargrams at different levels of processing. It proposes a novel method to denoise and despeckle raw radargrams by exploiting a generative network (diffusion probabilistic model) that models the noise statistical properties. The model is iteratively trained to learn the radargram information loss due to the noise level increment. When reversing the process, the network estimates the noise statistical properties, thus denoise unseen radargrams. To validate the effectiveness of the proposed method, we conducted successful experiments on raw Experiment Data Record (EDR)

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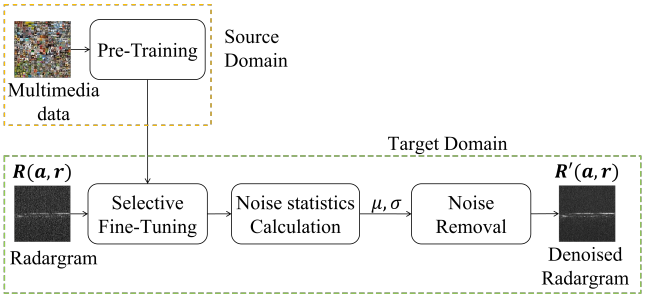


Fig. 1: Flowchart of the proposed denoising method.

radargrams of Mars obtained through the Shallow Radar (SHARAD) system.

2. PROPOSED DENOISING METHOD

Let's define a radargram $\mathbf{R}$ as a 2D matrix of N_S samples and N_T traces as

$$\mathbf{R} = \{R(r,a)|a \in [1, N_T], r \in [1, N_S]\}, \quad (1)$$

where $R(r,a)$ is the amplitude of the signal and contains reflections from the surface and subsurface (signal pixels) surrounded by the background (no-signal pixels). Given the non-ideality of the acquisition environment and sensors, radargrams are affected by several noisy contributions (e.g., ionosphere distortion, galactic noise, receiver thermal noise) characterized by different statistical distributions. Section 2.1 analyses the statistical distributions of the noisy contributions in radargrams at different levels of processing. The proposed method relies on this analysis and proposes a generative neural network (diffusion probabilistic model [8]) that learns the statistical distribution of the noise. The model is trained considering the forward and reverse processes that generate a Markov chain of diffusion steps. The forward process aims at learning the information loss in radargrams due to the increasing noise level. In contrast, during the reverse process, the network recovers the denoised radargram by estimating the noise statistical properties for each pixel in terms of mean and variance and removing it. When trained, the network can accurately estimate the statistics of different noise levels of unseen radargrams to generate noise maps that can be removed from the data. Finally, as domain adaptation demonstrated to reduce the computational time [5, 6], we propose to pre-train the network on a proxy task and then fine-tune it to denoise radargrams (see Fig. 1).

2.1. Radargram statistical analysis

Radargrams are affected by different types of noise (e.g., thermal, galactic, and speckle noises, ionospheric distortion) having different statistical properties. This paper focuses on the receiver noise, the cosmic noise, and the speckle. The

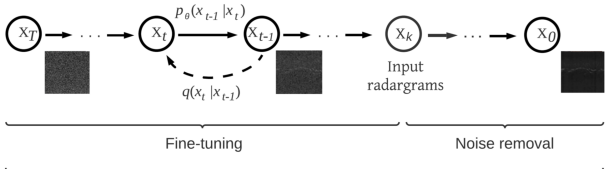


Fig. 2: Representation of the DDPM Markov chain.

receiving sensor introduces the thermal noise that follows a Rayleigh distribution, which typically models the amplitude oscillation of a zero-mean additive Gaussian noise (AWGN) [2]. The isotropic cosmic microwave background causes the galactic noise that can be approximated as Rayleigh distributed in the power domain [9]. The speckle is a noise-like phenomenon caused by the interference of complex elementary echoes backscattered by a surface roughness larger than the radar wavelength [10]. Speckle appears in radargram as a granular pattern and can be approximated in amplitude as multiplicative noise. [3] shows that the real and imaginary parts of the complex speckle components are independent and identically distributed (i.i.d.) and follow a Gaussian distribution in amplitude. Considering the different theoretical distributions of signal and noise components [11, 2, 3], we perform a statistical analysis of raw radargrams (without range compression, calibration, and SAR focusing) with the method in [2]. The analysis shows that the complex components (real and imaginary parts) of the raw data are independent and identically distributed (i.i.d.) and follow a Gaussian distribution like the SAR data in [11, 3]. Therefore, the proposed method will focus on processing the real and imaginary parts of the complex raw radargrams.

2.2. Noise characterization and removal

We propose to train a diffusion probabilistic model (DPM) [8] to denoise raw radargrams R. DPM aims to learn the decay of information due to the increasing noise level (forward process) so that the process can be reversed to recover the signal (reverse process). The forward diffusion process aims at generating a Markov chain of T diffusion steps that smoothly contaminates the input data $R_0 = R$ by injecting random Gaussian noise to get a sequence of noisy samples $\{R_1, \ldots, R_t, \ldots, R_T\}$ considering the transition probability $q(R_t|R_{t-1})$ (see Fig. 2). The noise is controlled by the mean and the variance values that depend on a hyperparameter $\{\beta_t \in (0,1)\}_{t=1}^{T}$ being defined as a constant or gradually changed for each step t. With the increasing t, the data lose the distinguishable spatial features to reach an isotropic Gaussian distribution with $T \to \infty$. The reverse diffusion process aims at learning the unknown reverse process transition probability $q(R_{t-1}|R_t)$ to estimate the denoised image $\hat{R}$ from Gaussian noise. However, unlike the forward process, estimating the previous state R_{t-1} available in the current state

R_t requires the knowledge of all the previous chain gradients, which requires learning a model that predicts such estimates. Therefore, the network is trained to estimate the probability $p_\theta(R_{t-1}|R_t)$ that is a parameterized model of $q(R_{t-1}|R_t)$ based on the network weights θ and the time step t. The network is trained by optimizing the negative log-likelihood $\mathcal{L}$ of the training data [8, 12].

$$\mathcal{L} = E_q \left(-\log(p(R_T)) - \sum_{t \geq 1} \log \frac{p_\theta(R_{t-1}|R_t)}{q(R_t|R_{t-1})} \right). \quad (2)$$

Authors in [8, 12] simplify $\mathcal{L}$ so that the network is trained to estimate the noise statistics in terms of mean and variance for each pixel (a, r) at each time step t. Finally, the denoised radargram $R'(a, r)$ is computed by removing the estimated noise from the input R_0. This is done by generating for each radargram a noise map defined for each pixel using the estimated mean and variance values. The noise map is then subtracted from the radargram to obtain the denoised radargram.

As domain adaptation dramatically reduces the computing time [6], we propose pre-training the network for all the diffusion steps $t \in [1, T]$ on a large multimedia dataset, such as ImageNet or Celeb. We fine-tune the network on a subset of time steps $t \in [T_n, T], T_n > 1$ with radargrams (see Fig. 2). At $t = T_n$, the network estimates the noise statistics to reconstruct the input radargram R_0. Once fine-tuned, the network predicts the noise statistics of the radargrams for the whole chain of diffusion steps till $t = 1$. The choice of T_n is critical as it identifies the noise level that will be removed from the radargram– the smaller the value of T_n, the lower the amount of noise removed from the radargram. With a greater value of T_n, more noise is removed from the radargram at the cost of more possible artifacts and loss of geometrical details.

3. EXPERIMENTAL RESULTS

We validated the proposed method on raw Experiment Data Record (EDR) radargrams of Mars acquired by the Shallow Radar (SHARAD). We considered all the radargrams acquired at the poles (latitude larger than $|75°|$) when SHARAD is in occultation to minimize the distortions due to the ionosphere. The radargrams are divided into 10798 patches of 256 traces and 256 samples. The patches are divided into 80% for the training (8638 patches) and 20% for the test phase (2160 patches). To have independent sets, we force the patches in the train set to belong to radargrams different from those for the test set. As a neural network, we adapt that in [8] to the RS data characteristics (e.g., one-dimensional channel), and we experimentally set the initial learning rate to 0.0005 and a weight decay to 0.9999. The training uses Adam optimizer and a batch size of 24. Finally, we experimentally set the pre-training noise step to $T = 1000$ and the fine-tuning so that $T_n \in [5, 75]$ with step 5. However, for space constraints, here we report the results for $T_n = 15$ and $T_n = 25$. For the quantitative evaluation, we calculated the Equivalent Number of Looks (ENL) on homogeneous parts and the Structural Similarity Index Measure (SSIM), Signal-to-Noise Ratio (SNR), and Peak Signal-to-Noise Ratio (PSNR) metrics.

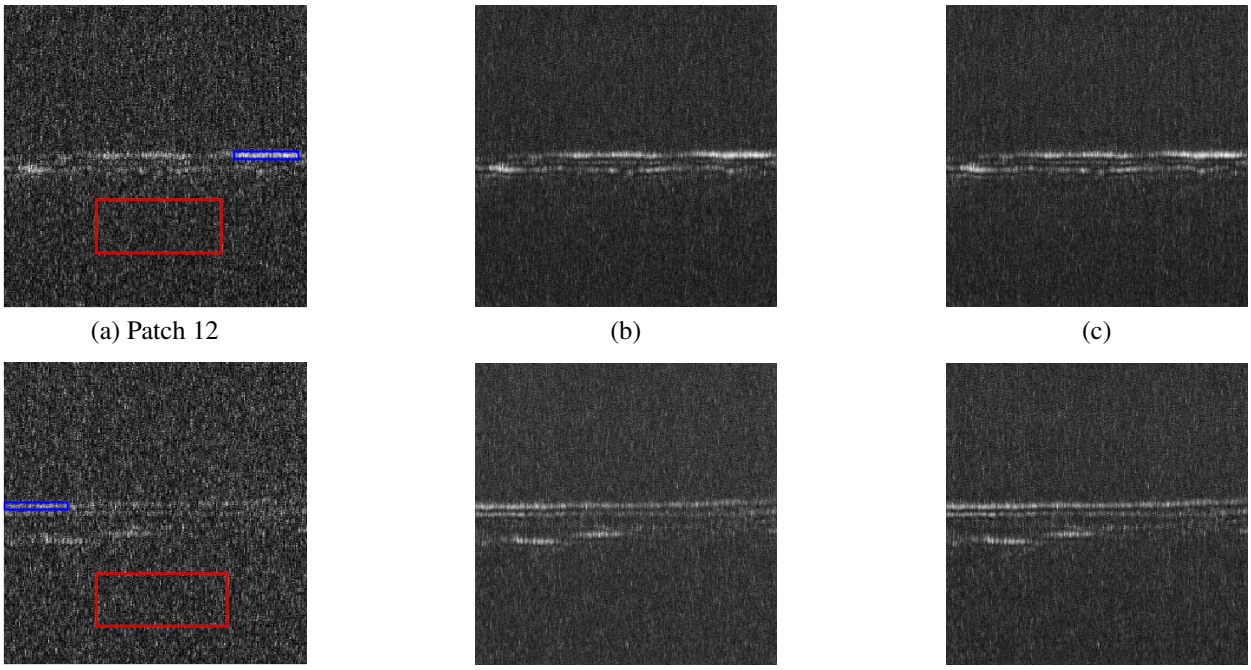
Table 1: ENL performance of the proposed method for the patches in Fig. 3.

Patch ID	T_n	ENL signal	ENL background
12	0	20.67	16.94
	15	23.88	33.01
	25	38.94	57.13
13	0	10.56	12.55
	15	15.75	25.38
	25	26.63	38.36

Table 2: Performance of the proposed method for the patches in Fig. 3 and the whole dataset (normalized radargrams).

Patch ID	T_n	SNR	PSNR	SSMI
12	0	5.35	10.34	1 (reference)
	25	6.16	12.36	0.824
13	0	4.58	11.72	1 (reference)
	25	6.45	13.02	0.743
All dataset	0	5.88	19.88	1 (reference)
	25	6.92	22.50	0.795

The qualitative results in Fig 3 show that the proposed method denoises the radargrams by reducing the speckle and increasing the homogeneity in the background. Moreover, the results show that increasing the time step (Fig. 3.c and f) from $T_n = 15$ to $T_n = 25$ further decreases the noise level and enhances the reflections. The quantitative results show that the proposed method increases the homogeneity of the background and the reflection by reducing noise. For the patch in Fig 3.a, the ENL is 10.67 and 16.94 for the signal (red rectangle in Fig 3.a) and the background (blue rectangle in Fig 3.a), respectively. The ENL improves to 23.88 and 33.01 and 38.94 and 57.13, respectively (Fig 3.b and c) (see Table 1). This trend is also visible for SNR, PSNR, and the SSMI metrics in Table 2– the metrics for $T_n = 25$ are higher than the reference values of the original patch. These preliminary results highlight the goodness of the proposed method for denoising radargrams. However, the results highlight the importance of choosing the value of T_n, i.e., the amount of noise to be removed. Increasing T_n decreases the amount of noise in the radargram until a certain point ($T_n = 30$ in our experiments); afterward, the method introduces artifacts. This highlights the importance of choosing T_n to fine-tune the network for denoising the radargrams.



(a) Patch 12 (b) (c)

(d) Patch 13 (e) (f)

Fig. 3: Experimental results on EDR SHARAD radargram patches. Fig (a) and (d) show the original patches, and Fig. (b-c) and (e-f) show the related denoised radargrams with time steps $T_n = 15$ and $T_n = 25$, respectively.

4. CONCLUSIONS

The paper proposed a deep-learning-based method for denoising radargrams in an unsupervised way by exploiting a generative model that learns the noise statistical properties. The generative model learns the information loss due to increasing the noise level. This process is reversed so that the network can estimate the noise statistical properties at each level and thus denoise unseen radargrams. We successfully validated the proposed method on SHARAD data. The results show that the proposed method can remove the noise at different levels depending on the parameter T_n. Choosing T_n is critical as a higher value removes more noise at the cost of introducing artifacts.

In future work, we plan to perform an extensive analysis considering data acquired in other scenarios (e.g., arid area of the Moon and Mars) and by other sounders with different parameters (e.g., LRS and MARSIS). Moreover, we plan to investigate the possibility of automatically estimating the optimum number of T_n.

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